Assignment: The "Release Manager" Roleplay

Course: SDLC & DevOps Fundamentals

Project: The "Smart-Attend" Mobile App

1. The Scenario

You are the Junior Project Manager for a university startup building a **Student Attendance App**. The app uses Geo-fencing to mark attendance automatically when a student enters the classroom.

The Pivot: Halfway through development, the Dean adds a new requirement: **Biometric (Face ID) Verification** must be integrated to prevent "proxy" attendance.

2. Task 1: The "Waterfall" Failure Analysis

Analyze why using a **Waterfall (Traditional) Model** for this project would cause a crisis when the new Biometric requirement is introduced. Identify **3 specific reasons** for failure, such as:

- **Rigidity:** The inability to go back to the requirement phase once development or testing has started.
- **Cost of Change:** High expenses and delays associated with changing the "Blueprint" mid-way.
- **Delayed Testing:** Discovering integration issues only at the very end of the 6-month cycle.

3. Task 2: The "Agile" Pivot (The Scrum Framework)

Switch to the **Scrum Framework** to handle this change. Break the project into **two 2-week Sprints**:

- **Sprint 1 (MVP):** Focus on the "Minimum Viable Product"—Basic UI, User Login, and the core Geo-fencing logic.
- **Sprint 2 (The Update):** Integrate the Biometric (Face ID) logic and refine the reporting dashboard based on early feedback.

4. Task 3: The Sprint Backlog (Practical Example)

Create a **Sprint Backlog** for **Sprint 1** in a spreadsheet. This should be a list of tasks derived from the "Product Backlog" that the team commits to finishing.

Sample Sprint Backlog:

Task ID	**User Story (As a..., I want..., So that...)**	**Priority**	**Estimated Hours**
US01	As a student, I want to login via university ID so I can access my profile.	High	4

Task ID	User Story (As a..., I want..., So that...)	Priority	Estimated Hours
US02	As a teacher, I want to view a real-time list of present students.	High	6
US03	As a system, I want to detect if a student is within 10m of the classroom.	High	8
US04	As a student, I want a notification when my attendance is marked.	Medium	3
US05	As an admin, I want to manually override attendance for technical errors.	Low	4

5. Submission Instructions

1. **Create a GitHub Repository:** Name it SDLC-Assignment-[YourName].
2. **Upload Files:** Include your Analysis (PDF) and your Backlog (Excel/CSV).
3. **Documentation:** Write a README.md explaining why **CI/CD** (Continuous Integration/Continuous Deployment) would be used to push these Sprint updates to students automatically.
4. **Deadline:** 14-03-2026